Refraction at Plane Surface

- Refraction

↳ The process of bending of light on passing from one medium to another medium is called Refraction.

↳ The refraction occurs due to change in velocity of light from one medium to another medium.

↳ The velocity of light is more in rarer medium than in denser medium.

↳ Light bends towards the normal when it passes through rarer to denser medium and light bends away from the normal when it passes through denser to rarer medium.

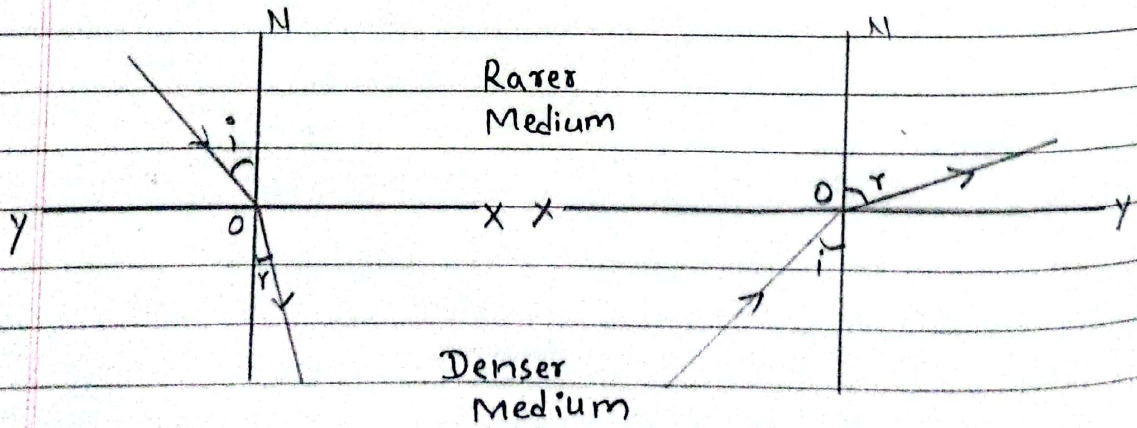


fig- Refraction of Light

- Laws of Refraction of Light

(i) Snell's law

↳ The ratio of sine of angle of incidence to the sine of angle of refraction is always constant for the given pair of medium, and that constant is called refractive index (μ).

$$\boxed{\text{i.e.} - \frac{\sin i}{\sin r} = {}_1\mu_2}$$

[Refractive index of second medium with respect to first medium]

(ii) Incident Ray, Refracted ray and normal all lie at a point of the same plane.

- Limitation of Snell's law

↳ Snell's law cannot be applied for the normally incident ray. At that condition,

$\mu = \frac{\sin 0}{\sin 0} = \frac{0}{0}$ = [indeterminant form]

- Absolute Refractive Index

↳ The ratio of Velocity of light at vacuum to the velocity of light at medium is called Absolute Refractive index.

i.e. - $\mu = \frac{\text{Velocity of light at vacuum (c)}}{\text{Velocity of light at medium (v)}}$

$\Rightarrow \boxed{\mu = \frac{c}{v}}$

$_a\mu_g = 1.5$ $[3/2]$
$_a\mu_w = 1.33$ $[4/3]$
$_a\mu_D = 2.4$
$_1\mu_1 = 1$

- Relative refractive index

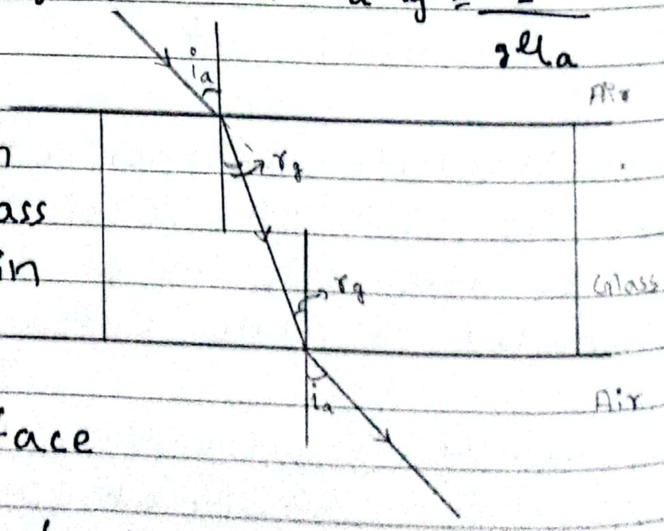
↳ It is define as the ratio of velocity of light in first medium to the velocity of light in second medium

i.e- $_1\mu_2 = \dfrac{\text{velocity of light in } 1^{st} \text{ medium}}{\text{Velocity of light in } 2^{nd} \text{ medium}}$

- Principle of reversibility of light

↳ When the ray of light suffers from number of reflection of refraction then it will returns from the same path if we reversed its final path.

Prove that : $_a\mu_g \times {}_g\mu_a = 1$ OR $_a\mu_g = \dfrac{1}{_g\mu_a}$



Consider a Ray of light passes from air to glass and glass to air as shown in figure.
Now,
For air-glass surface

Refractive index of glass with respect to air is given by

$$_a\mu_g = \frac{\sin i_a}{\sin r_g} \rightarrow (i) \quad \text{[From snell's law]}$$

Also,
For glass-air surface

Refractive index of air with respect to glass is given by

$$_g\mu_a = \frac{\sin r_g}{\sin i_a} \rightarrow (ii)$$

Multiplying eqn (i) & (ii)

$$_a\mu_g \times {_g\mu_a} = \frac{\sin i_a}{\sin r_g} \times \frac{\sin r_g}{\sin i_a}$$
$$= 1$$

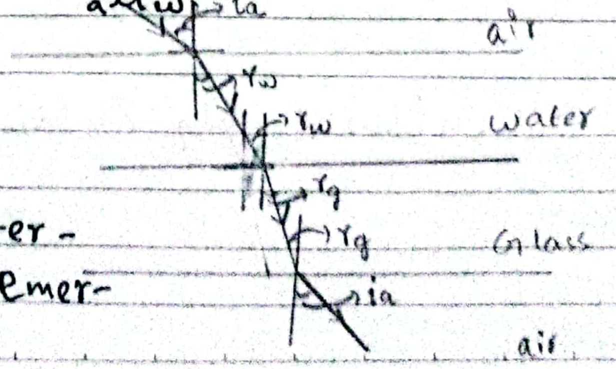
$$\boxed{\therefore {_a\mu_g} \times {_g\mu_a} = 1} \quad \text{or,} \quad \boxed{_a\mu_g = \frac{1}{_g\mu_a}}$$

Proved.

Prove that : $_a\mu_w \times {_w\mu_g} \times {_g\mu_a} = 1$

OR

$$_w\mu_g = \frac{_a\mu_g}{_a\mu_w}$$



Consider a ray of light passes from 3 consecutive medium i.e -air-water-glass and finally emerges to air medium.

as shown in figure.

Now, for air-water surface, Refractive index of water with respect to air is given by :-

$$ {}_a\mu_w = \frac{\sin i_a}{\sin r_w} \rightarrow (i) $$ [From snell's law]

Water glass medium
Refrective inden of glass with Respect to water is given by

$$ \therefore {}_w\mu_g = \frac{\sin r_w}{\sin r_g} \rightarrow (ii) $$

finally, For glass-air medium
Refrective inden of air with respective to glass.
i.e. $$ {}_g\mu_a = \frac{\sin r_g}{\sin i_a} \rightarrow (iii) $$

Multiplying eqn (i), (ii) & (iii)

$$ {}_a\mu_w \times {}_w\mu_g \times {}_g\mu_a = \frac{\sin i_a}{\sin r_w} \times \frac{\sin r_w}{\sin r_g} \times \frac{\sin r_g}{\sin i_a} $$
$$ = 1 $$

$$ \therefore {}_a\mu_w \times {}_w\mu_g \times {}_g\mu_a = 1 $$

Proved.

Again,

$${}_{w}\mu_{g} = \frac{1}{{}_{a}\mu_{w} \times {}_{g}\mu_{a}}$$

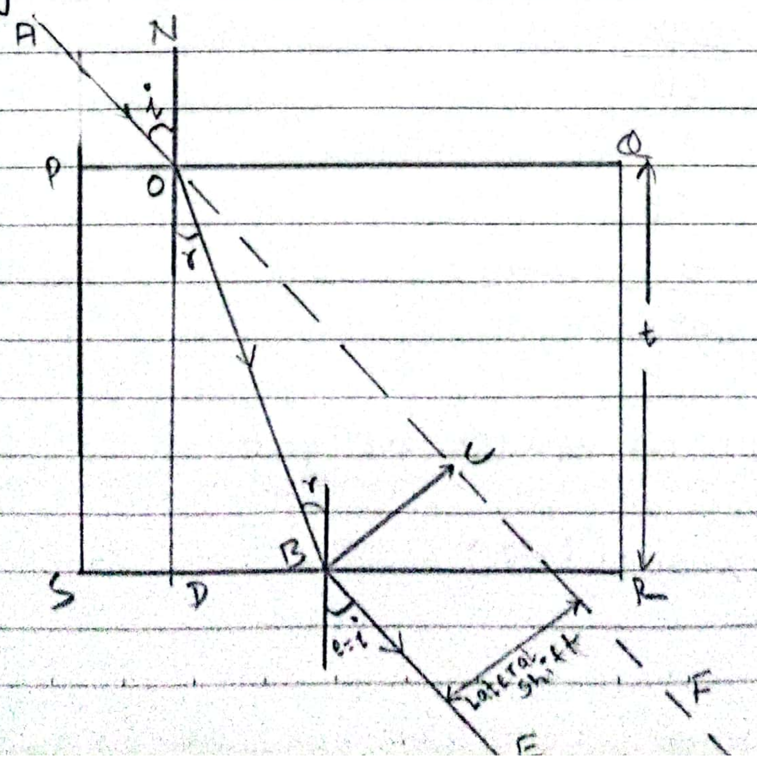
$${}_{w}\mu_{g} = \frac{{}_{a}\mu_{g}}{{}_{a}\mu_{w}} \quad \left[\because {}_{g}\mu_{a} = \frac{1}{{}_{a}\mu_{g}}\right]$$

$$\boxed{\therefore {}_{w}\mu_{g} = \frac{{}_{a}\mu_{g}}{{}_{a}\mu_{w}}}$$

Lateral shift

↳ When the ray of light passes from rectangular glass slab then the emergent ray and original direction of incident ray are parallel with each other.

↳ Lateral shift is the perpendicular distance between energent ray and original direction of incident ray.



Consider PQRS is a rectangular glass slab ~~w~~ with thickness 't'. A ray of Light AO is incident on the glass slab with angle of incidence 'i'. This ray is refracted along OB with angle of refraction 'r'. Finally, the ray of light emerges out along BE. AF is the original direction of incident ray. The perpendicular distance BC = P is the lateral shift.

Now from figure,

$\angle BOC = i - r$

~~Using~~ From $\triangle BOC$

$$\sin(i-r) = \frac{BC}{OB}$$

$$\sin(i-r) = \frac{P}{OB}$$

$$\Rightarrow P = OB \times \sin(i-r) \longrightarrow (i)$$

Again, from $\triangle DOB$

$$\cos r = \frac{OD}{OB}$$

$$\text{or, } \cos r = \frac{t}{OB}$$

$$\text{or, } OB = \frac{t}{\cos r} \longrightarrow (ii)$$

Using eqn (ii) on eqn (i) we get

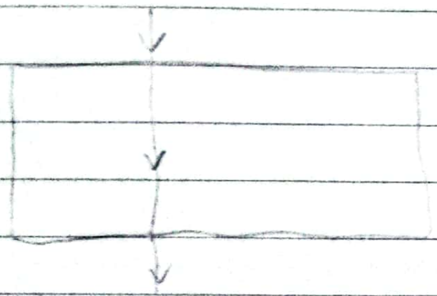
$$P = \frac{t}{\cos r} \times \sin(i-r)$$

$$\boxed{\therefore P = \frac{t \sin(i-r)}{\cos r}}$$

This is the required relation for lateral shift.

Special cases

- Case i - When $i=0$ (normal incidence

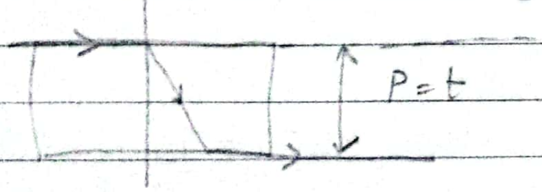


$$P = \frac{t \sin 0}{\cos 0}$$

$\Rightarrow P = 0$

This is the minimum value of lateral shift.

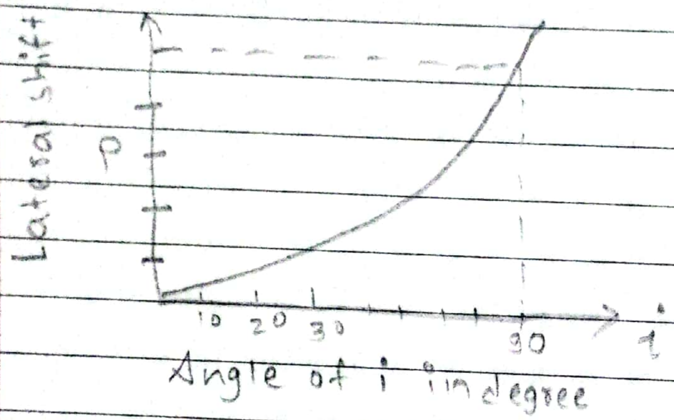
- Case ii - for $i = 90$ (grazing incidence)



$$P = \frac{t \sin(90-r)}{\cos r} = t$$

this ~~the~~ ~~m~~ maximum value of lateral shift.

Hence, on increasing the angle of incidence gradually from 0° to 90° the lateral shift also increases from a minimum value of zero to a maximum value equal to the thickness of the slab, which is represented by the following graph.

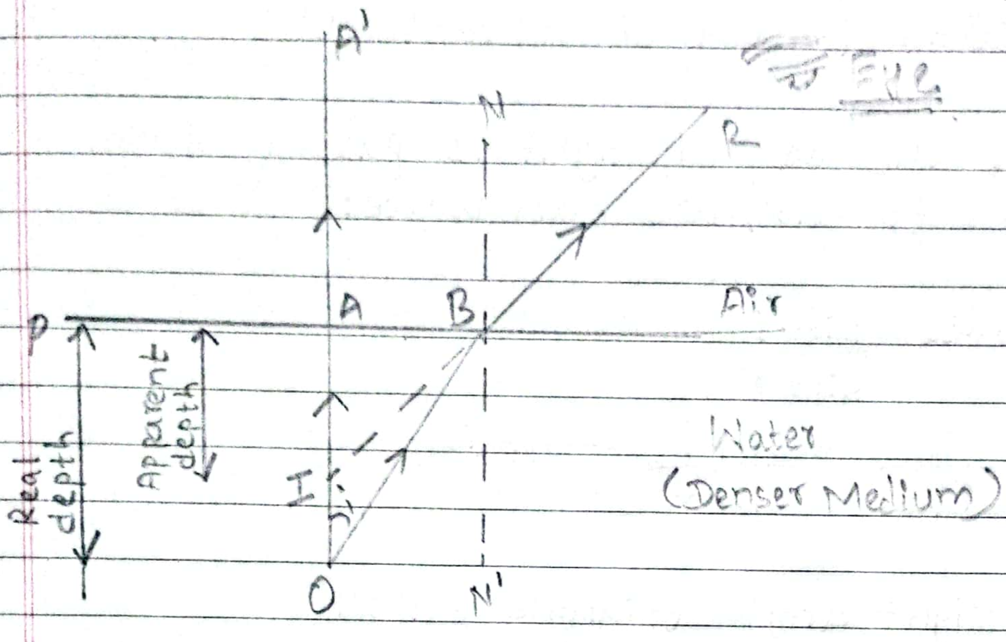


• Real Depth and Appearent depth

↳ When an object is observed from one medium to another medium, the object appears to be shifted from its original position.

The Actual depth of the object is called the real depth, and the observed depth is called apparent depth.

The real depth is greater than the apparent depth when viewed from a rarer to a denser medium. The apparent depth is greater than the real depth when viewed from a denser to a rarer medium.



Consider an object placed at the bottom of a beaker containg water up to a thickness 't'. A ray of light OA comes out normally without bending. Another ray, OB, is incident at point B with an angle of incidence i and refracts along BR, with an angle of refraction r.

The ray BR is produced backward to ~~a~~ point I, so I is the ~~di~~ position of the virtual image of the object at a distance d from the object.

In the figure:
AO = t = real depth
AI = apparent depth

$OI = d =$ displacement of object
$\angle AOB = i$ (Angle of incidence)
$\angle AIB = r$ (Angle of refraction)

Since the ray of light is passing from water to air, we can write:

$$ {}_w\mu_a = \frac{\sin i}{\sin r} $$

$$ \text{or, } \frac{1}{{}_a\mu_w} = \frac{\sin i}{\sin r} \longrightarrow (i) $$

In right angle triangle $\triangle BOA$

$$ \sin i = \frac{AB}{OB} $$

In right angle triangle $\triangle BIA$

$$ \sin r = \frac{AB}{IB} $$

Eqn (i) becomes

$$ \frac{1}{{}_a\mu_w} = \frac{\frac{AB}{OB}}{\frac{AB}{IB}} $$

$$ \text{or, } {}_a\mu_w = \frac{OB}{IB} $$

For small aperture of eye, the point A and B should be very close to each other.
So, $OB \approx OA$ and $IB \approx IA$

$$ \mu = \frac{OA}{IA} $$

$\therefore \mu = \dfrac{\text{real depth}}{\text{Apparent depth}}$

Also,

Displacement of the Object (d)
= Real depth (t) − apparent depth

or, $d = \text{Real depth} - \dfrac{\text{Real depth}(t)}{\mu}$

~~or.~~ $\boxed{d = t\left(1 - \dfrac{1}{\mu}\right)}$

• Critical Angle and Total Internal Reflection

↳ When a ray of light passes from a denser medium to a rarer medium, on increasing the angle of incidence, the angle of refraction also increases.

The angle of incidence in the denser medium for which the angle of refraction is 90° is called the critical angle. It is denoted by C.

If the angle of incidence is greater than the critical angle, then the light reflects totally into the denser medium. This phenomenon is called total internal reflection.

* Conditions for Total Internal Reflection:

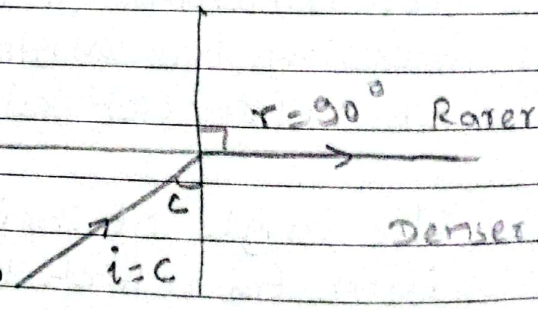
① A ray of light should pass from a denser medium to a rarer medium.

② The angle of incidence should be greater than the critical angle, i.e, $i > c$.

* Prove that: $\mu = \frac{1}{\sin c}$, where c is the critical angle.

↳ Consider a ray of light passing from water to air with an angle of incidence equal to the critical angle, i.e. $i = c$. In this case, the angle of refraction in air is $90°$ so $r = 90°$.

$$\cancel{{}_w\mu_a = \frac{\sin i}{\sin r}}$$



Since the ray is passing from water (denser) to air (rarer), we can write the refractive index of air with respect to water as:

$${}_w\mu_a = \frac{\sin i}{\sin r}$$

$$\text{or, } \frac{1}{{}_a\mu_w} = \frac{\sin c}{\sin 90}$$

$$\text{or, } {}_a\mu_w = \frac{1}{\sin c}$$

$$\therefore \mu = \frac{1}{\sin c}$$

Some examples of total Internal reflection:

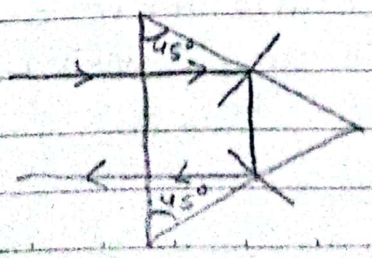
① Mirage

↳ Mirage is an optical illusion produced on desert during hot summer days.

The layer of air near the surface becomes very hot which acts as rarer medium. The other layers above it gradually cooler and acts as denser medium. The ray of light from denser medium suffers total internal reflection and mirage is noticed.

Total reflecting Prism

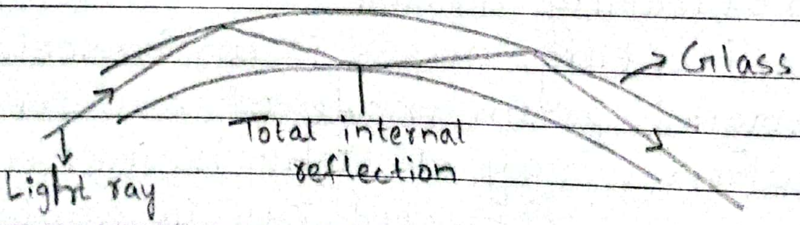
↳ Total reflecting prism is an isosceles right-angled prism made of crown glass. When a ray of light is incident on a face and reflected ray in another face makes the angle with normal greater than the critical angle, it will total internally reflected. Binoculars and submarine periscopes use such types of prisms.



Prism reflection is better than the mirror reflection because there will be total internal reflection from the prism where no loss of intensity takes place so the image will be very bright and clear but in case of mirror reflection, some of the light intensity is absorbed so image will be dim and unclear.

Optical Fibers

↳ It is very narrow glass tube from which light is passed from one end to another end due to total internal reflection. It is used to study inner part of our body (endoscopy, laparoscopy and Telecommunications.)



Brilliancy of diamond

↳ The refractive index of diamond is 2.4 and hence, its critical angle is small, 24.6°. Diamond ~~sir~~ shines because of multiple total internal reflections occurs in it due to its small critical angle and cutting.

Numerical and Figure Based Questions

1. If $\mu_g = 1.5$, $\mu_w = 1.33$ and $\mu_d = 2.82$

a) Find the refractive index of water with respect to glass?

→ Soln,

$${}_g\mu_w = \frac{\mu_w}{\mu_g}$$

$$= \frac{1.33}{1.5}$$

$$= 0.88$$

$\therefore {}_g\mu_w = 0.88$

b) Find the refractive index of water-glass interface.

→ Soln,

$${}_w\mu_g = \frac{\mu_g}{\mu_w}$$

$$= \frac{1.5}{1.33}$$

$$= 1.12$$

$\therefore {}_w\mu_g = 1.12$ #.

c) Refractive index of diamond when it completely immersed in water?

→ Soln,

$${}_w\mu_d = \frac{\mu_d}{\mu_w}$$

$$= \frac{2.82}{1.33}$$

$$= 2.12$$

$\therefore {}_w\mu_d = 2.12$ #.

2. Angle of incidence in water is 50°. Calculate the angle of refraction in glass.

→ Given,

$\angle i = 50°$

$\mu_w = 1.33$

$\mu_g = 1.5$

Now,

$$_w\mu_g = \frac{\mu_g}{\mu_w}$$

$$= \frac{1.5}{1.33}$$

$$= 1.13$$

Again,

$$_w\mu_g = \frac{\sin i}{\sin r}$$

or, $1.13 = \frac{\sin 50}{\sin r}$

Or, $\sin r = \frac{0.76}{1.13}$

Or, $\sin r = 0.67$

or, $r = \sin^{-1}(0.67$

$\therefore r = 42.6°$ //.

3. How long time will the light take to travel a distance of 500m in water? [Refractive inden for water is 1.33 and velocity of light in vacuum is 3×10^8 m/s].

→ Given,

$\mu_w = 1.33$

Velocity of light in vacuum (c) $= 3 \times 10^8$

Velocity of light in water (v) = ?

$\mu_w = \frac{c}{v}$

or, $v = \frac{3 \times 10^8}{1.33}$

or, $v = 2.2556 \times 10^8$ m/s.

Again,

Velocity (v) = $\frac{\text{distance}}{\text{time taken}}$

or, $2.25 \times 10^8 = \frac{500}{T}$

or, $T = \frac{500}{2.25 \times 10^8}$

or, $T = 2.21 \times 10^{-6}$ s

$\therefore T = 2.21 \times 10^{-6}$ s

4. Find the lateral shift produced by a glass slab of thickness 8 m and refractive index 1.5, when a ray incident on the glass slab with angle of incident 40°.

→ Given,

$t = 8$ m

$\mu = 1.5$

$\angle i = 40°$

$P = ?$

We know that,

$\mu = \frac{\sin i}{\sin r}$

or, $1.5 = \frac{\sin 40}{\sin r}$

or, $\sin r = \frac{0.64}{1.5}$

or, $\sin r = 0.4285$

$\therefore r = 25.37°$

Again,

Lateral shift (P) $= \frac{t \sin(i-r)}{\cos r}$

$= \frac{8 \sin(40-25.37)}{\cos 25.37}$

$= 8 \frac{\sin 14.62}{\cos 25.37}$

$= 8 \times \frac{0.2525}{0.90}$

$= 8 \times 0.279$

$= 2.23$ m

$\therefore$ Lateral shift (P) $= 2.23$ m $_{Ans.}$

5. A microscope is focused on a scratch on the bottom of a beaker. Turpentine is poured into the beaker to a depth of 4 cm, and it is found necessary to raise the microscope through a vertical distance of 1.25 cm to bring the scratch again into focus. Find the refractive inden of turpentine.

→ Soln,

Real depth (R.D) = 4 cm
Apparent depth (A.D) = 4 - 1.25 cm
= 2.75 cm

$$\mu = \frac{\text{Real depth}}{\text{Apparent depth}}$$
$$= \frac{4 \text{ cm}}{2.75 \text{ cm}}$$
$$= 1.45$$

$\therefore$ Refractive index (μ) = 1.45

6. What is the apparent position of an object below a rectangular block of glass 6 cm thick if a layer of water 4 cm thick is on the top of the glass? (Refractive index of glass = $\frac{3}{2}$ and of water = $\frac{4}{3}$)

→ Soln,

For glass

$t_g = 6$ cm

$\mu = \frac{3}{2}$

$d_g = ?$

$$d_g = t_g\left(1 - \frac{1}{\mu_g}\right)$$
$$= 6\left(1 - \frac{1}{3/2}\right)$$
$$= 2 \text{ cm}$$

For water

$t_w = 4$ cm

$\mu = \frac{4}{3}$

$d_w = ?$

$$d_w = t_w\left(1 - \frac{1}{4/3}\right)$$
$$= 4\left(1 - \frac{1}{4/3}\right)$$
$$= 1 \text{ cm}$$

$\therefore d = d_g + d_w$
$= 2 \text{ cm} + 1 \text{ cm} = 3 \text{ cm}$ above bottom

7. A transparent cube of 12 cm edge contains a small air bubble. Its apparent depth when viewed through one face of the cube is 6 cm and when viewed through opposite face is 2 cm. What is the actual distance of bubble from the first face?

→ Soln,

For 1^{st} face

Real Depth (R.D) = x (let)

Apparent depth (A.D) = 6cm

$$\mu = \frac{\text{Real Depth}}{\text{Apparent depth}}$$

$$= \frac{x}{6} \longrightarrow (i)$$

For Opposite face

R.D = $12 - x$

A.D = 2 cm

$$\mu = \frac{R.D}{A.D}$$

$$= \frac{12-x}{2} \longrightarrow (ii)$$

From eqn (i) & (ii)

$$\frac{x}{\cancel{6}\,3} = \frac{12-x}{\cancel{2}}$$

$x = 36 - 3x$

$4x = 36$

$x = 9$

$\therefore x = 9$ cm

$\therefore$ Real depth = 9 cm.

8. Calculate the Critical angles of (i) glass-water and (ii) water-air interfaces if object lies in the denser medium.

(i) glass-water

$${}_{w}\mu_{g} = \frac{\mu_g}{\mu_w}$$

$$\Rightarrow {}_{w}\mu_{g} = \frac{1.5}{1.33}$$

$$\Rightarrow \frac{1}{\sin c} = 1.127$$

$$\Rightarrow \sin c = \frac{1}{1.127}$$

$$\Rightarrow c = \sin^{-1}\left(\frac{1}{1.127}\right)$$

$$\Rightarrow c = 62.45^\circ$$

$$\therefore c = 62.45^\circ$$

ii) Water-air interfaces

$${}_{a}\mu_{w} = \frac{\mu_w}{\mu_a}$$

$$\Rightarrow {}_{a}\mu_{w} = \frac{1.33}{1.00}$$

$$\Rightarrow \frac{1}{\sin c} = 1.33$$

$$\Rightarrow \sin c = \frac{1}{1.33}$$

$$\Rightarrow c = \sin^{-1}\left(\frac{1}{1.33}\right)$$

$$\Rightarrow c = 48.75$$

$$\therefore c = 48.75^\circ$$

9. An optical fiber ($\mu = 1.72$) is surrounded by a glass coating ($\mu = 1.50$). Find the critical angle for total internal reflection at fibre-glass interface.

→ Given,

$\mu_o = 1.72$

$\mu_g = 1.5$

We know that,

$${}_g\mu_o = \frac{\mu_o}{\mu_g}$$

$$\Rightarrow {}_g\mu_o = \frac{1.72}{1.5}$$

$$\Rightarrow \frac{1}{\sin c} = 1.14$$

$$\Rightarrow \sin c = \frac{1}{1.14}$$

$$\Rightarrow c = \sin^{-1}\left(\frac{1}{1.14}\right)$$

$$\Rightarrow c = 60.70$$

$\therefore c = 60.70°$ #.

10. Light from a luminous point on the lower face of a rectangular glass slab, 2 cm, thick, strikes the upper face and totally reflected rays outline a circle of radius 3.2 cm on the lower face. What is refractive index of the glass?

→ Given,

Glass slab thickness $(t) = 2$ cm

Circle radius on lower face $(r) = 3.2$ cm

From snell's law,

$$\sin\theta_c = \frac{1}{\mu}$$

~~or, $\tan\theta_c = \frac{1}{\sqrt{\mu^2-1}}$~~

$$\tan\theta = \frac{p}{b} = \frac{1.6}{2}$$

$$\text{or, } \theta = \tan^{-1}\left(\frac{1.6}{2}\right)$$

$$= 38.65^\circ$$

Again,

$$\mu = \frac{1}{\sin c}$$

$$= \frac{1}{\sin 38.65^\circ}$$

$$= 1.60$$

$\therefore$ Refractive index $= 1.6$

• Conceptual Questions.

1. Why does the bottom of a swimming pool always appear shallower then it actually is?

→ Because of refraction of light. When light rays from the bottom of the pool pass from water to air, they bend away from the normal, making the bottom appear higher than it really is.

2. Why does the sun look slightly oval when it is near the horizon?

→ Light coming from the sun reaches to the earth through the atmosphere which has unequal density of air and the refractive inden is different for different layers. When a ray ~~so~~ from the lower part of sun reaches to the earth surface, it suffers more refraction than the ray from the top of sun. So, the lower portion of the sun is shifted to apparent position up and the vertical diameter appears shorter than horizontal diameter. Hence the sun looks a little oval when it is at the horizon. It is due to unequal atomospheric refraction.

3. A ray of light ~~f~~ in air strikes a glass surface. Is there a range of angles for which total internal reflection takes place?

→ No, Total internal Reflection does not Occur when light travels from air to glass because it happens only from a denser to a rarer medium.

• Exam pattern Questions.

1) Why does diamond sparkle?
→ Diamond sparkles because of total internal reflection. When light enters a diamond, it bends and reflects many times inside due to its high refractive index and well-cut surfaces, before coming out. This repeated reflection and dispersion of light produce a bright sparkle.

2) Why do in summer roads often appear to be covered with water when seen from a distance? Explain.
→ In summer, the air near the road is very hot and the air above it is cooler. Light from the sky bends (refracts) as it passes through these layers of air with different densities. The ray bends upwards after reflection from the hot surface, making it look like they come from a pool of water. This illusion is called a mirage.

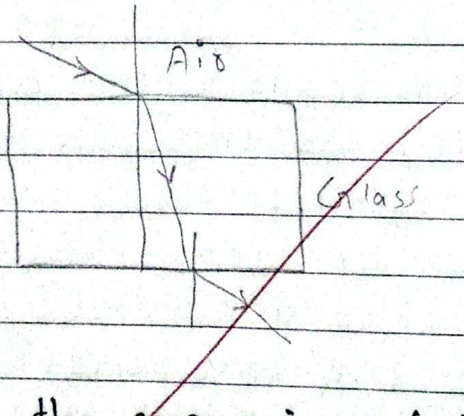
3. A glass slab is placed over a page in which letters are printed in different colors. Will the image of all letters lie in same plane?

→ No, the image of all letters will not lie in the same plane because the refractive index of glass is different for different colors. Light of each color bends differently, so their images are formed at slightly different depth inside the glass slab.

• Numerical and figure Based Questions.

1. •

a) Complete the path of the ray in the figure shown to show lateral shift.



b) Write the expression of Lateral shift as seen in the figure above.

→ Expression for lateral shift

$$(P) = \frac{t \sin(i-r)}{\cos r}$$ Where,

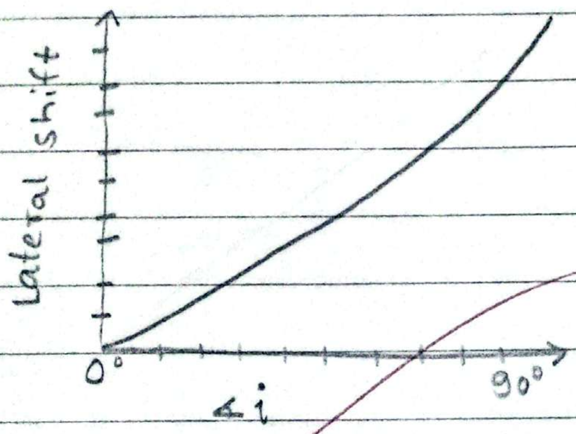
t = thickness of the glass slab

i = angle of incidence

r = angle of refraction

c) Sketch a group to explain how the lateral shift varies with angle of incidence.

→ When the ray is incident normally ($i = 0°$), lateral shift (P) is equal to 0. As the angle of incidence increases, the lateral shift increases and becomes maximum just before total internal reflection. The lateral shift (P) increases with the angle of incidence (i).



d) For what angle of incidence, the lateral shift produced by a parallel sides glass slab is zero?

→ When the angle of incidence is 0°, the lateral shift produced by a parallel sides glass slab is zero.